

Omnidirectional Amphibious Locomotion via Internal Mass Actuation

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Abstract—Field robots must traverse varied terrain and obstacles while remaining robust to water, debris, vegetation, and physical contact. We present MARBLE, a fully enclosed omnidirectional amphibious rolling robot driven entirely by internal mass redistribution. Three mutually orthogonal linear sliders shift internal masses to generate body rotation, while an orientation-aware controller maps planar velocity commands into slider positions. A rigid spherical shell encloses all active mechanisms and simultaneously serves as the terrestrial contact surface, buoyant enclosure, and mounting structure for passive fins that enable water-surface propulsion. Rotation of the same shell architecture hence produces rolling on land and surface propulsion in water without mechanical reconfiguration or separate locomotion actuators. The spherical morphology further allows the robot to accommodate changes in body orientation and contact location during direct interactions with terrain and obstacles. We evaluate MARBLE through omnidirectional locomotion characterization, traversal across heterogeneous terrestrial environments, aquatic surface locomotion, land-water transitions, and deliberate obstacle interactions. These experiments demonstrate how a single enclosed mechanical architecture can combine omnidirectional mobility, cross-medium locomotion, and tolerance to environmental contact. MARBLE provides a compact design for field mobility across heterogeneous terrain, obstacles, and land-water transitions. We will open-source all software and hardware design. Our website is <https://generalroboticslab.com/MARBLE>

I. INTRODUCTION

Field robots operate in environments where terrain, obstacles, and environmental conditions can vary substantially over the course of a mission. Inspection, monitoring, and disaster response may require traversal across uneven ground, vegetation, debris, and aquatic regions [1]–[3]. These environments challenge both locomotion and the mechanical structures that produce it. Wheels, limbs, propellers, and other exposed moving components can become obstructed, contaminated, or entangled by vegetation, debris, and water [4]–[6]. Robust field mobility depends on the ability to move across heterogeneous environments while tolerating the physical contact and environmental exposure encountered during traversal.

Multimodal robots extend mobility by combining locomotion capabilities for different environments. Air-ground platforms transition between flight and terrestrial motion [7]–[9], while amphibious robots integrate terrestrial and aquatic locomotion [10], [11]. Bio-inspired designs further demonstrate how mechanical structures can support multiple

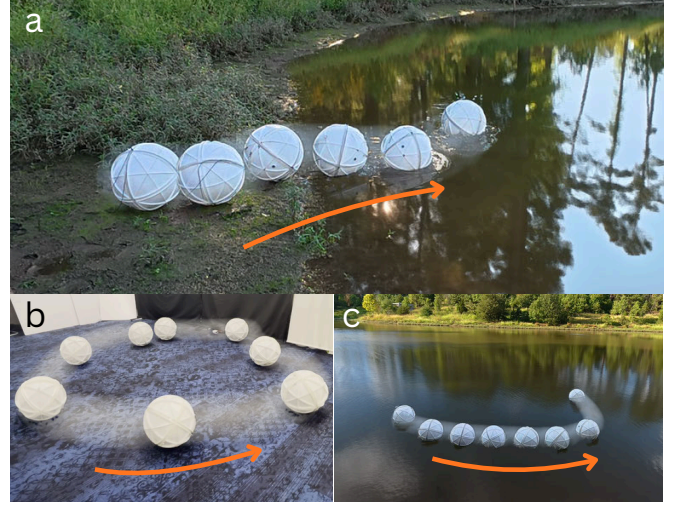


Fig. 1. Time-lapse images of MARBLE showing (a) land-to-water entry, (b) terrestrial rolling, and (c) water-surface propulsion. The same internal mass-actuation mechanism produces motion on land and water without mechanical reconfiguration.

locomotion modes. Prior robots have repurposed wings for walking, or coordinated limbs with body motion for both swimming and terrestrial locomotion [12], [13]. These designs substantially expand the environments accessible to a single robot. However, multiple locomotion modes can introduce additional mechanisms, exposed appendages, or mechanical reconfiguration. Each added mechanism creates another interface between the robot and its surroundings, where water, vegetation, debris, and physical contact can interfere with locomotion.

Spherical morphology offers a way to couple locomotion with mechanical robustness. A spherical body can roll through continuously changing contact locations without relying on a fixed upright orientation, allowing the exterior of the robot to serve directly as its locomotion interface. Spherical, symmetric, and tensegrity robots have used related geometric [14]–[18] or dynamic properties [19] for omnidirectional motion, impact tolerance, and locomotion across irregular terrain. Internal actuation can further keep motors and transmissions away from direct environmental contact [20]–[24]. Combining these properties creates an opportunity for cross-medium locomotion using a common body and actuation system across different environments.

We present MARBLE (Fig. 1), a fully enclosed omnidirectional amphibious rolling robot driven entirely by internal mass redistribution. Three mutually orthogonal linear sliders

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shift internal masses to displace the center of mass and generate body rotation. An orientation-aware controller maps planar velocity commands to slider positions as the body continuously reorients. All active components are enclosed within a rigid spherical shell. The shell serves as the rolling surface on land, provides buoyancy in water, and supports passive fins that convert body rotation into water-surface propulsion. The same internal actuation and rotating shell produce both terrestrial and aquatic locomotion without mechanical reconfiguration or separate propulsion actuators. During obstacle interactions, the spherical morphology accommodates changes in body orientation and contact location while the active mechanisms remain enclosed within the shell.

We evaluate MARBLE across terrestrial, aquatic, and obstacle-rich environments. Experiments characterize the directional response of the three-slider actuation system and evaluate locomotion across heterogeneous terrestrial surfaces, water-surface propulsion, land-water transitions, and deliberate obstacle interactions. Our experiments examine how a shared enclosed actuation design can provide omnidirectional mobility across different environments while maintaining mobility during environmental contact. Our main contributions are:

- We present a fully enclosed spherical robot design that uses a shared internal mass-actuation mechanism for omnidirectional terrestrial and aquatic locomotion.
- We design an orientation-aware controller that coordinates three orthogonal linear mass sliders for omnidirectional rolling under continuous body reorientation.
- We experimentally characterize terrestrial rolling, water-surface propulsion, land-to-water transitions, and object interaction through direct shell contact.

II. RELATED WORK

Multimodal and Amphibious Locomotion. Multimodal robots extend mobility across environments by reusing structures or actuation across locomotion modes. Air-ground robots combine flight with cage-supported rolling [7], [25], repurpose wings for terrestrial locomotion [12], or reconfigure appendages for wheeled, legged, and aerial motion [8]. Land-water robots similarly use shared appendages [10], transformable leg-flipper mechanisms [11], or coordinated spine-limb gaits [13] to move across terrestrial and aquatic environments. These systems demonstrate how shared mechanical structures or mechanical reconfigurations can reduce the hardware dedicated to individual locomotion modes for multimodal robots. Our MARBLE robot extends this idea to a fully enclosed design without reconfiguration. A single internal mass-actuation mechanism rotates the body on both land and water, with the enclosing shell serving as the terrestrial contact surface, buoyant enclosure, and support for passive aquatic propulsion.

Spherical Morphology and Terrain Interaction. Spherical and highly symmetric robots use distributed contact and body reorientation to maintain mobility across changing orien-

tations and contact conditions. Early multilimbed designs generate rolling through repeated body contacts [14], while spherical robots with radial telescopic legs actively vary their shape and contact configuration during locomotion [15], [16]. Tensegrity robots use structural compliance to absorb impacts and traverse irregular terrain [17], [18]. More recently, Argus further uses dynamic symmetry and radially distributed actuation to achieve nearly isotropic dynamic capabilities across different body orientations [19]. These designs show how robot morphology can accommodate changing orientations and contacts during locomotion. MARBLE uses a rigid spherical shell as a continuous external interface for terrain, water, and obstacle contact while keeping its active mechanisms enclosed within the body.

Internally Actuated Spherical Robots. Internally actuated spherical robots generate locomotion by moving mechanisms inside an outer shell. Prior designs use internal rotors to induce rolling and spinning [20], radial mass redistribution [21], tetrahedrally arranged moving masses [22], and multiple pendulums [23]. Other designs use dedicated drive and steering mechanisms [26] or omnidirectional wheels acting against the inner surface of the shell [27]. Close recent work has extended internal actuation to amphibious locomotion by combining pendulum-driven rolling, rotator-based steering, and passive fins for aquatic propulsion [28]. In contrast, MARBLE uses three orthogonal linear mass sliders for orientation-aware omnidirectional locomotion. The same internal actuation drives both terrestrial rolling and water-surface propulsion, avoiding separate steering and propulsion mechanisms while keeping all active locomotion components enclosed.

III. MECHANICAL DESIGN AND LOCOMOTION

A. Shell Design

A rigid spherical shell encloses the structural frame, actuators, and electronics (Fig. 2(d)) and forms the primary interface between MARBLE and its surroundings. The assembled shell has an outer diameter of 387 mm and a mass of 1.05 kg. Symmetrically distributed passive fins extend 8 mm beyond the surface. The shell provides the rolling surface for terrestrial locomotion, while the fins generate hydrodynamic forces as the body rotates at the water surface.

The shell is fabricated from polylactic acid (PLA) using fused deposition modeling (FDM). Each hemisphere consists of four identical printed segments. A radial silicone O-ring seals the interface between the two hemispheres, which are secured together by two cast silicone retaining bands with Shore A hardness 10 (Smooth-On Dragon Skin 10). Three plastic bags surrounding the internal components provide secondary protection against water ingress.

The sealed shell also provides buoyancy for water-surface locomotion. The shell encloses a nominal internal volume of 0.030 m^3 , with the equilibrium draft determined by the balance between the total robot weight and the buoyant force generated by the displaced external volume. Because the permeability of FDM-fabricated parts depends on print

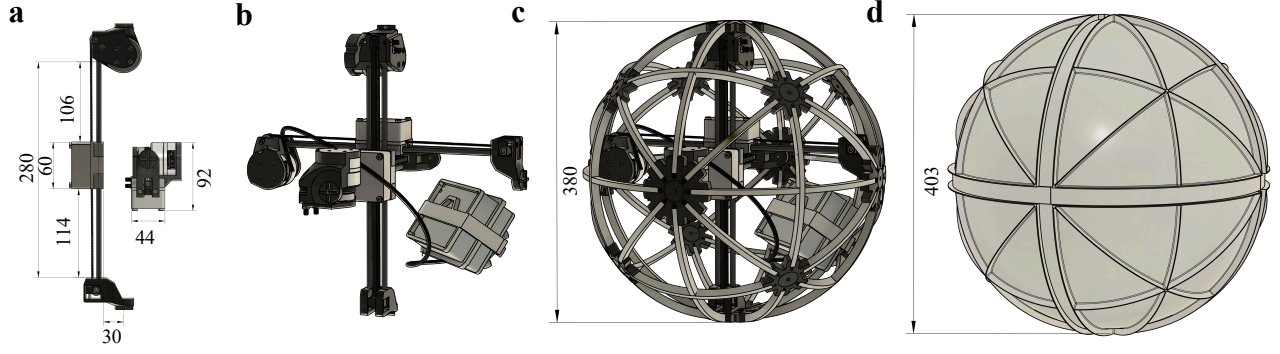


Fig. 2. **Mechanical design of MARBLE.** (a) Linear mass-slider module. (b) Internal configuration with three orthogonal sliders and the electronics capsule. (c) Internal frame without the outer shell. (d) Complete robot with the sealed shell and passive fins.

and assembly parameters, we characterize the sealing performance experimentally to find the submerged depth of MARBLE. Our design allows the same exterior structure to serve as the terrestrial rolling surface, buoyant enclosure, and passive aquatic propulsion interface.

B. Structural Frame

The internal frame holds the three slider modules and onboard electronics (Fig. 2(b, c)). It consists of 48 beams and 14 hubs, with three beam variants connecting six primary hubs and eight secondary hubs through snap-fit joints. The snap-fit construction eliminates 96 fasteners compared with an assembly using two fasteners per beam. Secondary hubs and connecting beams fill the gaps between the primary structural members and produce a more continuous rolling external profile. This frame provides direct access to the actuators and electronics during development and testing. All field experiments use the complete sealed shell.

C. Mass-Slider Actuation

Locomotion is generated by three identical linear mass sliders aligned with mutually orthogonal body axes (Fig. 2(a,b)). Each module translates a 700 g mass over a total stroke of 220 mm. A GT2 timing belt drives the mass along carbon-fiber guide rods with igus iglidur linear bearings. The three rails are offset by 30 mm to provide clearance between the moving masses.

Each slider is driven by a CubeMars GL40 II motor under joint-level proportional-derivative (PD) position control at 100 Hz. An additional support bearing reduces cantilever loading on the drive assembly under belt tension. By coordinating the slider positions, MARBLE can shift its center of mass along arbitrary directions within the reachable mass-redistribution workspace. The nominal center of mass depends on the rail offsets and the distribution of the fixed components when all three sliders are centered. The sliders operate at a nominal 16 volts from a 4S lithium polymer battery. The motors draw up to 1.88 A, have a peak torque of 0.68 N m and peak speed of 145.4 rad s⁻¹.

D. Locomotion Principle

Let the body frame B be fixed to the shell with its origin at the geometric center. The center of mass expressed in B

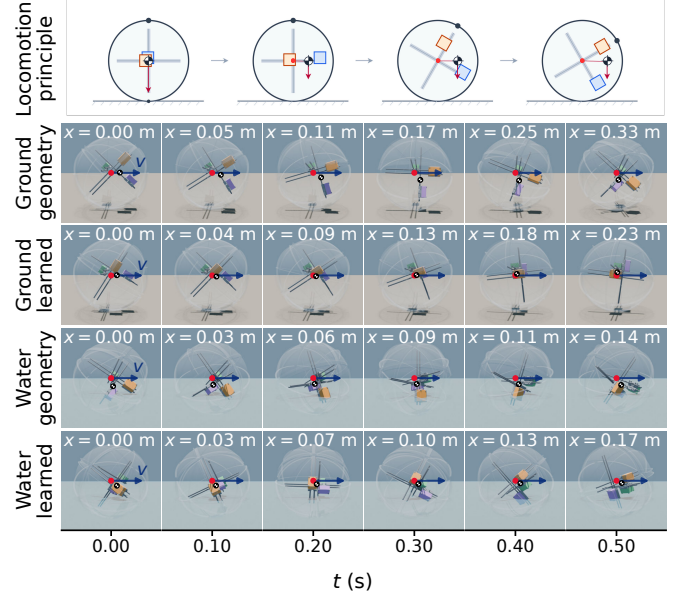


Fig. 3. **Locomotion principle of MARBLE.** Top: displacing the internal slider masses offsets the center of gravity from the shell center, and the resulting moment about the ground contact rolls the shell. Below: simulation of ground and water-surface locomotion at 0.5 m s⁻¹ using geometric and learned controllers, sampled at the same instants. The transparent shell shows the three sliding masses, colored one per rail. The quartered circle denotes their center of gravity, the red dot the shell center, and the blue arrow the heading direction.

is

$$\mathbf{r}_G^B = \mathbf{r}_{G,0}^B + \frac{1}{M} \sum_{i=1}^3 m_i q_i \mathbf{e}_i^B, \quad (1)$$

where $\mathbf{r}_{G,0}^B \in \mathbb{R}^3$ is the nominal center of mass with the sliders centered and M is the total robot mass. For each slider $i \in \{1, 2, 3\}$, $m_i \in \mathbb{R}_{>0}$ denotes the moving mass, $q_i \in \mathbb{R}$ its displacement from the centered position, and $\mathbf{e}_i^B \in \mathbb{R}^3$ is the corresponding unit rail direction and translation axis. The three axes are mutually orthogonal.

On land, mass redistribution creates a horizontal offset between the center of mass and the ground contact. Gravity then produces a moment that rotates the body. For a horizontal lever arm d , the gravitational moment has magnitude $\tau_g = Mgd$. With sufficient tangential traction, the shell rolls without slip. As the body rotates, the controller continuously

updates the slider positions to maintain the center-of-mass bias toward the commanded direction. This simplified model captures the primary gravitational mechanism and neglects transient inertial effects, slip, and discrete contacts introduced by the fins.

On the water's surface, the same mass redistribution offsets the center of mass from the line of action of buoyancy and generates body rotation. The submerged fins experience hydrodynamic forces as the shell rotates. Fins above the waterline encounter substantially lower resistance, producing an asymmetric interaction over the rotating shell and net surface propulsion [28]. Aquatic motion uses the same three active sliders as terrestrial locomotion. The change in locomotion mode arises from the interaction between the rotating shell and the surrounding medium. The resulting thrust depends on the immersion depth, fin geometry, and body rotation rate and is characterized experimentally.

E. Sensors and Computing

Onboard computation is provided by an Orange Pi Zero 3W. The computer interfaces over USB with a TM171 nine-axis inertial measurement unit (IMU) and a Seeed Studio XIAO nRF52840 Sense Plus microcontroller. The microcontroller communicates with the three motor drives through an MCP2515-based CAN interface. The IMU estimates body orientation for transforming motion commands into the rotating body frame. For experimental evaluation, robot position and velocity are measured externally from video.

IV. CONTROL

We develop two types of controllers with a common low-level actuation interface. An orientation-aware geometric controller directly maps planar motion commands to slider positions using the measured body orientation. A learned controller predicts the same three slider position targets from onboard observations. Both controllers use the joint-level position control. Figure 3 shows the resulting mass redistribution for both controllers during simulated terrestrial and aquatic locomotion.

A. Coordinate Frames and Command Mapping

We define a magnetic north–west–up world frame W , a body frame B centered on the robot with axes aligned with the three sliders, an IMU frame S , and an operator command frame C . The rotation matrix $\mathbf{R}_{AB} \in \text{SO}(3) \subset \mathbb{R}^{3 \times 3}$ maps vectors expressed in frame B into frame A . The body orientation is computed using the measured IMU orientation and the fixed IMU mounting transform,

$$\begin{aligned} \mathbf{R}_{WB} &= \mathbf{R}_{WS} \mathbf{R}_{BS}^\top, \\ \mathbf{R}_{WS} &= \text{diag}(1, -1, -1) \mathbf{R}_{W_{\text{NED}}S}, \end{aligned} \quad (2)$$

where W_{NED} denotes the IMU's magnetic north–east–down reference frame and $\text{diag}(1, -1, -1) \in \text{SO}(3)$ converts it to the world-frame convention.

The command frame C provides a fixed heading for operator input. Its orientation differs from W by a constant

yaw offset $\psi_0 \in [-\pi, \pi)$:

$$\mathbf{R}_{WC} = \begin{bmatrix} \cos \psi_0 & -\sin \psi_0 & 0 \\ \sin \psi_0 & \cos \psi_0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in \text{SO}(3). \quad (3)$$

The operator sets ψ_0 once per run.

For the geometric controller, the planar velocity input $\mathbf{v}_{xy}^c = [v_x^c, v_y^c]^\top \in \mathbb{R}^2$ specifies the desired direction of travel. A nonzero input defines a virtual target at a fixed distance δ from the robot center,

$$\mathbf{p}^C = \begin{cases} \frac{\delta}{\|\mathbf{v}_{xy}^c\|} \begin{bmatrix} v_x^c \\ v_y^c \\ 0 \end{bmatrix}, & \|\mathbf{v}_{xy}^c\| > 0, \\ \mathbf{0}_3, & \|\mathbf{v}_{xy}^c\| = 0, \end{cases} \quad (4)$$

$\delta = 1.694 \text{ m}.$

The normalization removes the command magnitude, so this controller uses a binary stop or move command with a continuous planar direction. The target expressed in the rotating body frame is

$$\mathbf{p}^B = \mathbf{R}_{WB}^\top \mathbf{R}_{WC} \mathbf{p}^C. \quad (5)$$

B. Geometric Controller

Each slider i moves along a line in the body frame,

$$\ell_i(q_i) = \mathbf{a}_i^B + q_i \mathbf{e}_i^B, \quad i \in \{1, 2, 3\}, \quad (6)$$

where $\mathbf{a}_i^B$ is the mass position at $q_i = 0$ and $\mathbf{e}_i^B$ is the corresponding unit rail direction. The rail axes satisfy

$$(\mathbf{e}_i^B)^\top \mathbf{e}_j^B = 0, \quad i \neq j. \quad (7)$$

We project the virtual target onto each slider axis to obtain the desired mass displacement,

$$q_i^* = (\mathbf{e}_i^B)^\top (\mathbf{p}^B - \mathbf{a}_i^B). \quad (8)$$

This projection selects the point along each rail closest to the common virtual target. As the robot rotates, $\mathbf{p}^B$ changes continuously and redistributes the three masses to maintain the commanded world-relative direction. The commanded displacement is clipped to the calibrated travel range,

$$q_i^{\text{cmd}} = \text{clip}(q_i^*, q_{i\min}, q_{i\max}). \quad (9)$$

Here, $\text{clip}(x, a, b) = \min\{\max\{x, a\}, b\}$ for $x, a, b \in \mathbb{R}$ with $a \leq b$. A 220 mm stroke provides nominal limits of +114 mm, −106 mm, before applying a ± 10 mm safety margin. Zero motion command returns the masses to their centered configuration.

The slider displacement is converted to a motor angle setpoint through the belt transmission,

$$\theta_i = \theta_{i,0} + \sigma_i \frac{2\pi}{N_p p_{\text{GT2}}} q_i^{\text{cmd}}, \quad (10)$$

where $\theta_{i,0} \in \mathbb{R}$ is the encoder angle calibrated at the rail center and $\sigma_i \in \{-1, +1\}$ accounts for transmission and encoder polarity. Each transmission uses a pulley with $N_p = 50 \in \mathbb{Z}_{>0}$ teeth and a GT2 belt with pitch $p_{\text{GT2}} = 2 \text{ mm}$. The displacement q_i^{cmd} and belt pitch p_{GT2} are expressed

in the same length units. The final angle setpoints are transmitted to the three motor drives over CAN.

Overall, the geometric controller computes slider position setpoints by projecting a commanded virtual target onto each rail (Fig. 3). At each update, the measured body orientation is used to transform the target into body coordinates. The projected displacements are then clipped to the sliders' travel limits and converted to motor commands, redistributing the internal masses to create a rolling motion.

C. Learned Controller

We additionally train learned controllers for terrestrial and aquatic locomotion with reinforcement learning. Both policies share the same hardware interface of the geometric controller and output three slider position targets. This common action space allows the controllers to be exchanged at runtime without changes to the low-level motor control.

Simulation. Training uses 4096 parallel MuJoCo [29], [30] environments on a single GPU, with a 5 ms physics timestep and a 50 Hz control rate. The robot mass and inertia are derived from the CAD assembly. For ground contact, we approximate the shell by a sphere of radius 0.202 m, corresponding to the fin crests on which the robot rolls.

Water surface. Aquatic training requires modeling buoyancy and the interaction between the rotating fins and the water surface. MuJoCo's built-in fluid model assumes a uniform fluid, so it does not provide buoyancy at the water surface. We augment the MuJoCo dynamics by implementing a surface model that applies a force and a torque to the shell at every physics step, comprising buoyancy, the moment produced by the offset between the centers of mass and buoyancy, drag on the submerged shell, and thrust from the fins. The water is a flat surface at a fixed height z_w , with no waves or current. The sealed shell displaces water as a sphere of radius $R = 0.194$ m, smaller than the 0.202 m sphere used for ground contact, because water passes between the protruding fins instead of being displaced by them. The fins contribute only through the thrust term.

The buoyant force acts upward at the centroid of the submerged spherical cap, with magnitude $\rho_w g V_{\text{sub}}$, where ρ_w is the water density, g the gravitational acceleration, and $V_{\text{sub}} = \pi h^2(3R - h)/3$ the cap volume at submergence depth h . The sliders displace the center of mass horizontally while the center of buoyancy stays on the vertical line through the cap centroid, and this offset produces the moment that drives shell rotation in water. The submerged shell also experiences a quadratic drag force that opposes its velocity and scales with the submerged frontal area, for which we use the smooth-sphere drag coefficient of 0.47.

During water locomotion the robot is only partially submerged. Thrust comes from the fins below the waterline. We compute this thrust element-wise, discretizing the fins into short segments and treating each as a slender rod that resists flow across its axis $\hat{\mathbf{t}}_i$ and slips along it. The segments are arranged on 13 rings distributed over the shell, positioned so that every rotation axis has rings that load, since a ring

rotating about its own normal slides along itself and does not generate force. These rings do not carry mass, inertia, or collision geometry and only mark where the drag law is evaluated, so the simulated body remains the same as derived from CAD. Each ring is a tube seated on the outside of the shell, with radius $r_t = 8$ mm equal to the fin height. Dividing each ring into 24 segments resolves the waterline to 15° of arc. Only the component of the segment velocity $\mathbf{v}_i$ normal to the segment axis contributes, and expressing all quantities in the world frame,

$$\begin{aligned} \mathbf{F}_i &= -g_i^{\text{wet}} g_i^{\text{vent}} \frac{1}{2} \rho_w C_r A_s \|\mathbf{v}_{\perp i}\| \mathbf{v}_{\perp i}, \\ \mathbf{v}_{\perp i} &= \mathbf{v}_i - (\mathbf{v}_i \cdot \hat{\mathbf{t}}_i) \hat{\mathbf{t}}_i. \end{aligned} \quad (11)$$

Here $C_r = 1.0$ is the cross-flow drag coefficient of a circular cylinder, and $A_s = 2r_t \ell_s$ is the projected area of a segment of arc length ℓ_s . The segment velocity $\mathbf{v}_i$ follows from the rigid-body motion of the shell about its center of mass. Each $\mathbf{F}_i$ acts at the center of its segment. Summing the 312 segment contributions gives the thrust and the accompanying torque on the shell.

The wetting factor g_i^{wet} determines which segments are in the water and is therefore what produces the thrust, since the forces on opposite sides of a fully loaded shell would cancel as it rotates. It rises smoothly from zero to one as a segment submerges over one tube diameter, following $g_i^{\text{wet}} = u_i^2(3 - 2u_i)$ with $u_i = \text{clip}((z_w - c_{iz})/2r_t, 0, 1)$ for a segment center at height c_{iz} . A hard switch at the waterline would instead apply force steps on every revolution, which destabilizes training. The ventilation factor g_i^{vent} accounts for the air entrained behind a fast-moving segment, which reduces its drag. We model it as $g_i^{\text{vent}} = g_{\min} + (1 - g_{\min})/[1 + (\text{Fr}_i/\text{Fr}_c)^2]$, decaying from one toward $g_{\min} = 0.3$ as the segment Froude number $\text{Fr}_i = \|\mathbf{v}_{\perp i}\|/\sqrt{g h}$ passes the onset value $\text{Fr}_c = 2$. This number compares the segment speed with the wave speed set by the submergence depth h . The factor decays from one toward $g_{\min} = 0.3$ above $\text{Fr}_c = 2$, the value at which air begins to enter the wake.

The model omits added mass, wave-making drag, and interactions between individual fin wakes. The unmodeled effects can reduce the thrust generated by the robot, so we account for their combined effect using a per-episode scaling of the ring forces (Table I). The terrestrial and aquatic configurations share the robot model, actuation, observations, and action space. The aquatic configuration adds the water-surface forces and thrust randomization and uses a weaker travel-limit penalty.

Policy. The policy maps a 23-dimensional onboard observation to three slider position targets. Each action is scaled by 0.104 m about the centered rest pose to cover the usable travel in (9), and tracked by the same joint-level position controller used by the geometric controller. The observation contains body angular velocity, the command in operator frame C , slider positions and velocities, the previous action, and the body rotation matrix $\mathbf{R}_{WB}$. The rotation matrix provides a unique, continuous representation across all shell orientations, avoiding Euler-angle singularities and quater-

nion sign ambiguity. Three consecutive observation frames are stacked and encoded. During training, the critic additionally receives uncorrupted observations and the ground-truth linear velocity. The ground-truth linear velocity is unavailable to the deployed policy and provides privileged information for learning an internal velocity estimate from onboard observation.

Training. Velocity commands are sampled per axis within ± 0.7 m/s and resampled every 2 s to 8 s. Ten percent of the sampled commands are zero to include stationary behavior during training. The reward encourages planar velocity tracking while regularizing slider velocity, travel-limit violations, action rate, and action acceleration (Table I(a)). The tracking kernel widens with command speed to accommodate errors across the range from rest to 0.7 m/s. We anneal the action-rate penalty over the first 3000 iterations to avoid suppressing slider motion before a rolling gait develops.

Episodes last 20 s, without failure termination because the spherical body does not have a required upright orientation. Shell contact is part of normal locomotion. Each policy is trained for 15000 iterations using FlashSAC [31], an off-policy soft actor-critic algorithm with a distributional critic. We randomize contact, actuation, mass, and latency parameters during training as summarized in Table I(b).

Deployment. The learned policy runs onboard at 50 Hz and directly outputs the three slider position targets. Deployment uses the policy trained in simulation without hardware fine-tuning or an additional adaptation phase. The geometric and learned controllers differ only in how the slider targets are generated and share the same sensing, communication, and low-level actuation stack.

V. EXPERIMENTS

A. Terrestrial, Aquatic, and Transitional Locomotion

We first evaluated whether the shared internal actuation mechanism supported locomotion across terrestrial and aquatic environments. Fig. 4 shows representative time-lapse sequences and reconstructed planar trajectories on land and at the water surface. During terrestrial locomotion, MARBLE traveled 10.60 m over 14.00 s, reaching a mean speed of 0.745 ± 0.163 m/s and a maximum speed of 1.025 m/s, while following an approximately closed spatial trajectory. On water, MARBLE traveled 5.24 m over 14.00 s, with a mean speed of 0.375 ± 0.059 m/s and a maximum speed of 0.488 m/s. These results demonstrate sustained terrestrial and aquatic locomotion without introducing a separate aquatic propulsion mechanism.

We next commanded MARBLE to enter the water directly from land. The robot progressed continuously from terrestrial rolling to water-surface propulsion without mechanical reconfiguration or switching propulsion hardware. During the transition, it traveled 5.21 m over 23.93 s with a mean speed of 0.215 ± 0.146 m/s and a maximum speed of 0.840 m/s. The reduced mean speed occurred during shore-line entry, where the robot encountered changing contact and buoyancy conditions. The complete sequence shows

TABLE I: Reward terms and domain randomization shared by the terrestrial and aquatic policies unless otherwise noted.

(a) Reward terms		
Term	Expression r_i	w_i
Velocity tracking	$\exp(-(\ \mathbf{v}_{xy}^c - \mathbf{v}_{xy}\ ^2 + \lambda_z v_z^2)/\beta^2)$	+2.0
Slider velocity	$\sum_j \dot{q}_j^2$	-0.002
Travel limit	$\sum_j [(q_j - q_j^{\max})_+^2 + (q_j^{\min} - q_j)_+^2]$	-1.0
Action rate	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ ^2$	-0.05 $\rightarrow$ -0.2
Action acceleration	$\ \mathbf{a}_t - 2\mathbf{a}_{t-1} + \mathbf{a}_{t-2}\ ^2$	-0.05
Total reward $\sum_i w_i r_i$; $a \rightarrow b$ is a curriculum. $\mathbf{v}_{xy}, v_z$: shell velocity; $\mathbf{v}_{xy}^c$: command; $\beta = \max(0.5\ \mathbf{v}_{xy}^c\ , 0.3 \text{ m/s})$: kernel width; $\lambda_z = 0.1$; q_j : slider position; $q_j^{\max}, q_j^{\min}$: soft travel limits; $\mathbf{a}_t$: action; $(x)_+ = \max(x, 0)$.		
(b) Domain randomization and observation noise		
Parameter	Range	When
shell-ground friction	0.2 to 0.6 (abs)	S
Slider PD gains k_p, k_d	$\times 0.8$ to 1.2	R
Slider force limit	31.4 N to 37.7 N	R
Rail Coulomb friction	1 N to 7 N	R
Slider joint damping	0 to 0.01 (add)	R
Body mass	$\times 0.85$ to 1.15	9.6 s
Shell CoM offset	± 10 mm (add)	9.6 s
Command delay (to motors)	20 ms to 80 ms	20 ms
Sensor delay (to policy)	0 ms to 40 ms	120 ms
Ring thrust scale ^b	$\times 0.6$ to 1.0	R
Angular velocity	$\mathcal{N}(0, 0.1)$	step
Rotation matrix	$\mathcal{U}(\pm 0.05)$	step
Slider velocity	$\mathcal{U}(\pm 0.05)$	step
Slider position	$\mathcal{U}(\pm 5 \text{ mm})$, bias	step, R

When: once per environment (S), once per episode (R), or at the stated interval. Rail friction ranges from nearly free up to the 6.95 N that holds one slider against gravity, and the command delay brackets the 26 ms to 56 ms CAN round trip measured on hardware. ^bAquatic only.

continuous progression from terrestrial rolling to sustained aquatic locomotion using the same internal mass actuation.

B. Omnidirectional Velocity Characterization

We then characterized MARBLE's planar mobility across different travel directions and compared the geometric controller (GC) with the learned controller (LC). Commands were specified in the operator frame and transformed into the rotating body frame using the measured orientation, as described in Sec. IV.

Fig. 5 presents segmented velocity responses collected across different commanded headings. The distribution of velocity vectors over the full 360° range demonstrated that MARBLE's actuation mechanism can produce planar motion across arbitrary directions without a preferred fixed heading. This capability arose from coordinated mass redistribution along the three mutually orthogonal body axes.

We further compared the two controllers during water-surface locomotion. The learned controller reached a mean speed of 0.348 ± 0.114 m/s and a maximum speed of 0.592 m/s. The geometric controller reached a mean speed of 0.252 ± 0.088 m/s and a maximum speed of 0.461 m/s. The learned controller recording covered 94.10 m over 270.37 s, while the geometric controller covered 63.67 m over 252.20 s. These results show that both controllers produced omnidirectional aquatic locomotion, with the learned controller reaching higher translational speeds.

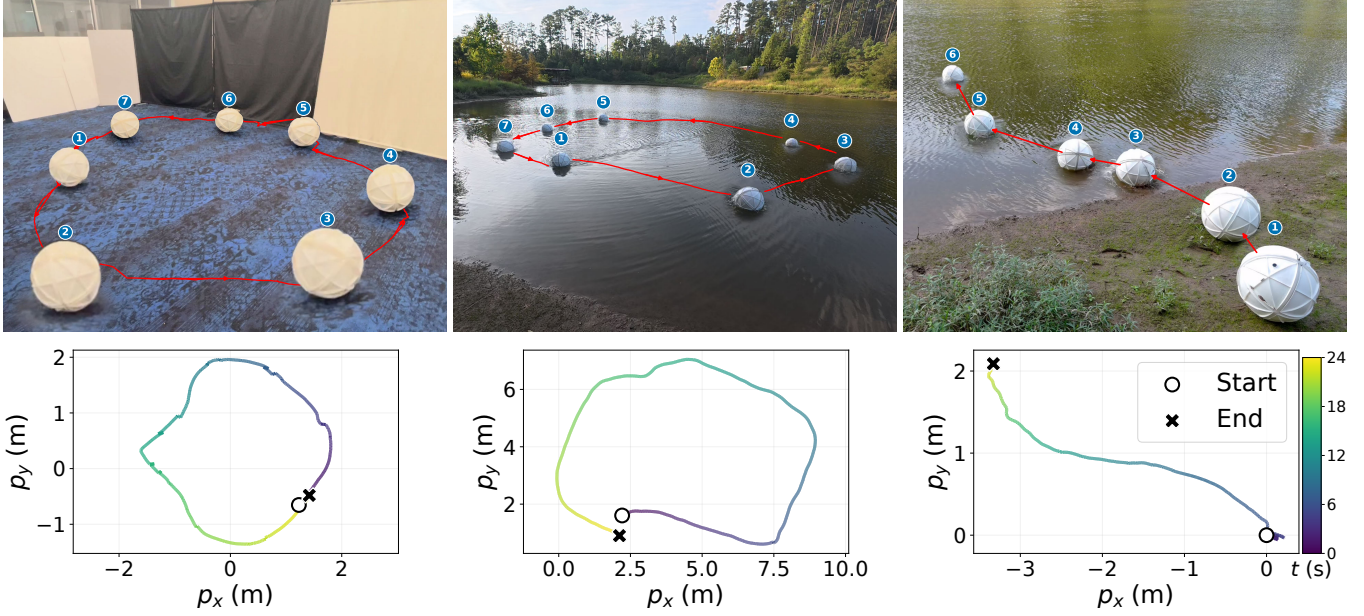


Fig. 4. Terrestrial, aquatic, and transitional locomotion of MARBLE. Top row: representative time-lapse sequences. Bottom row: corresponding planar position trajectories for (a) terrestrial, (b) aquatic, and (c) transitional locomotion. Trajectory color indicates temporal progression, while the circle and cross mark the start and end positions, respectively.

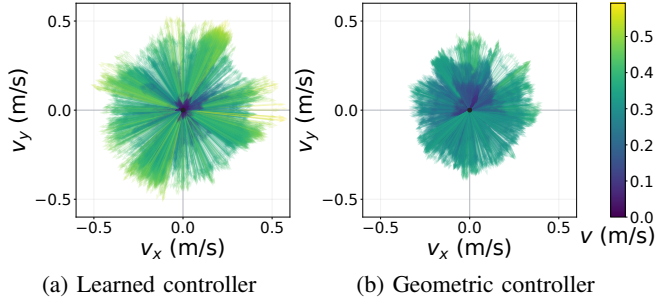


Fig. 5. Measured velocity vectors under joystick commands for (a) the learned controller and (b) the geometric controller. Velocities are estimated by numerical differentiation of camera-relative trajectories and plotted from a common origin. Each panel contains 7,568 samples with identical axis and color scales.

C. Object Interaction with Direct Shell Contact

The spherical morphology of MARBLE is designed to accommodate environmental contact without exposing the active locomotion mechanisms. We evaluated this property by commanding MARBLE to continuously push a buoy and allowing the shell and passive fins to contact the obstacle, as shown in Fig. 6.

During contact, the buoy changed the instantaneous contact location and perturbed the orientation and motion of the robot. MARBLE continued actuating its internal masses throughout the interaction and subsequently disengaged from the buoy and resumed locomotion. The active sliders, motors, and transmissions remain enclosed within the shell throughout the interaction. Fig. 6 shows the progression from approach through shell contact and subsequent recovery. These experiments illustrate how the spherical exterior of MARBLE can serve as the contact interface during obstacle interaction while the locomotion mechanism remains internal.

VI. CONCLUSIONS, LIMITATIONS, AND FUTURE WORK

We presented MARBLE, a fully enclosed amphibious robot that uses three orthogonal internal mass sliders for omnidirectional locomotion across land and water. The same actuation mechanism rotates the body for terrestrial rolling and drives passive fins for water-surface propulsion, without mechanical reconfiguration or separate active aquatic propulsion. Experiments characterized omnidirectional locomotion with geometric and learned controllers and demonstrated terrestrial rolling, water-surface propulsion, land-water transitions, and continued pushing during obstacle contact. Our results show how a shared internal actuation design can support cross-medium mobility while keeping the active locomotion mechanisms enclosed within the robot body.

The current platform is limited to water-surface locomotion, and its learned aquatic controller relies on an approximate hydrodynamic model. Though our learned controller performs adaptive locomotion, autonomous navigation is not supported. Future work will extend MARBLE toward autonomous field operation through additional onboard sensing and perception. Further development of the aquatic platform will investigate submerged locomotion with improved hydrodynamic modeling. These extensions can expand the same enclosed, mass-actuated design toward adaptive operation in more complex environments.

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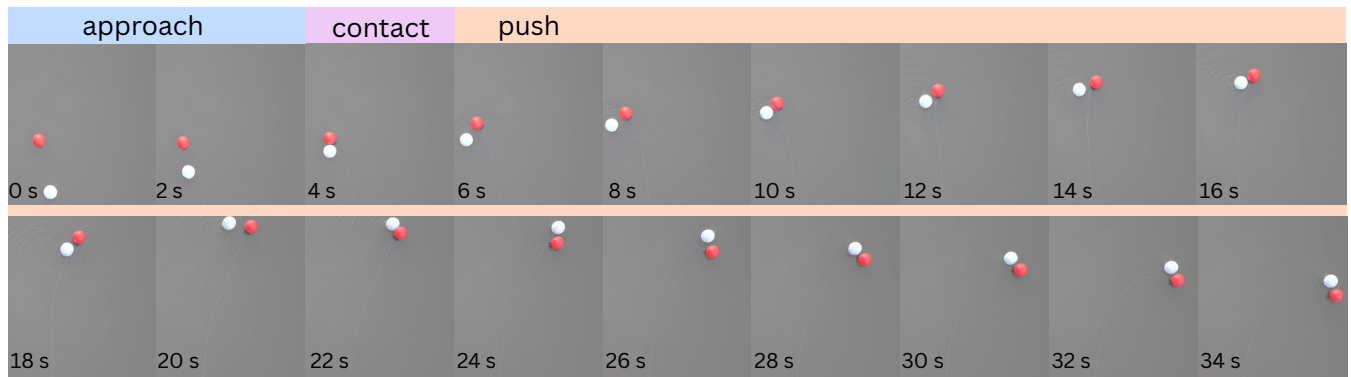


Fig. 6. Representative time-lapse sequence of obstacle interaction showing approach, direct shell contact, and continuous interaction with the buoy (red).

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